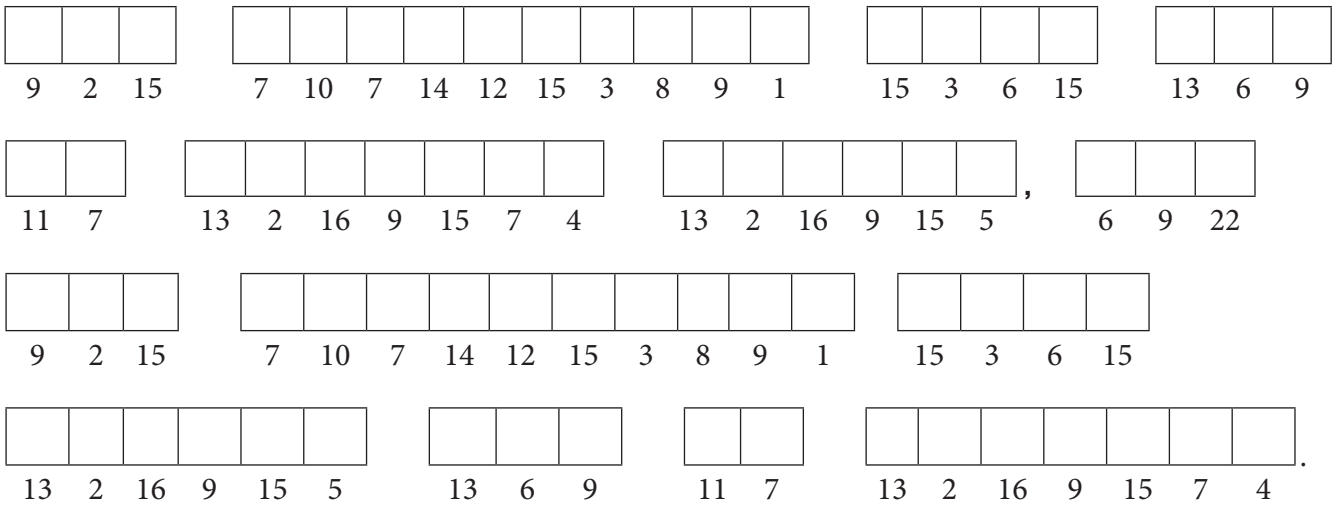
WORKSHEET

Reflections



Albert Einstein

Questions

Express the following reflections across the given lines through the origin as matrices:

1 The line with slope 3

2 The line with inclination 60°

3 The line whose equation is $x + y = 0$

4 The line with slope -2

5 The line with inclination $\frac{5\pi}{3}$

6 The line whose equation is $y = 2x$

7 The line with slope 1

8 The line with inclination $-30°$

9 The line whose equation is $3x + y = 0$

Find the image of each of the following points after reflection across the given line.

10 $(5, 7)$ across the line with slope 0

11 $(6, -1)$ across the line with inclination $\frac{2\pi}{3}$

12 $(-4, -3)$ across the line $x - y = 0$

Find the image of each of the following geometric shapes after reflection across the given line, to two decimal places where necessary.

13 $A(2, 4)$, $B(-4, 2)$, $C(1, -3)$ across the line with slope -1

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14 $A(-2, -3)$, $B(3, -5)$, $C(5, 3)$ across the line with inclination 45°

15 $A(1, -5)$, $B(2, 3)$, $C(-4, 1)$ across the line with inclination $\frac{\pi}{6}$

16 $A(0, 4)$, $B(4, 5)$, $C(-3, -5)$ across the line $2x + y = 0$

Solutions

A $\begin{bmatrix} -0.6 & 0.8 \\ 0.8 & 0.6 \end{bmatrix}$

B $\left(\frac{\sqrt{3} - 6}{2}, \frac{-1 - 6\sqrt{3}}{2}\right)$

C $A'(-4, -2)$, $B'(-2, 4)$, $C'(3, -1)$

D $\begin{bmatrix} -0.6 & -0.8 \\ -0.8 & 0.6 \end{bmatrix}$

E $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

G $\begin{bmatrix} -0.8 & 0.6 \\ 0.6 & 0.8 \end{bmatrix}$

H $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

I $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$

N $\begin{bmatrix} -0.8 & -0.6 \\ -0.6 & 0.8 \end{bmatrix}$

O $\begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

R $A'(-3, -2)$, $B'(-5, 3)$, $C'(3, 5)$,

S $\begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

T $A'(-3.83, 3.37)$, $B'(3.60, 0.23)$, $C'(-1.13, -3.96)$

U $A'(-3.2, 2.4)$, $B'(-6.4, -0.2)$, $C'(5.8, -0.6)$,

V $(5, -7)$

Y $(-4, -3)$

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